Haorui (Rae) Wang

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Portfolio | GitHub | LinkedIn

Education

University of Michigan – Ann Arbor Aug 2023 – May 2027

- Bachelor of Science in Data Science & Bachelor of Science in Statistics GPA: 3.7/4.0 (University Honors)

Technical Skills

Programming: Python, C++, SQL, R, Java, TypeScript
Machine Learning: PyTorch, scikit-learn, XGBoost, NLP, LLMs, Survival Analysis
AI Systems: Multi-Agent Systems, LLM Pipelines, Prompt Engineering, Tools APIs, Deep Learning, Transformers
Data: pandas, NumPy, PySpark, Matplotlib
Cloud: AWS, Azure, Microsoft Fabric, High Performance Computing
Certifications: IBM Machine Learning; Google Data Analytics; Google Business Intelligent

Technical and Professional Experience

Research Assistant — LLM-Based Clinical Extraction & Survival Modeling - UofM Jan 2026 – Present

- Built a production-style local LLM generator & evaluator extraction pipeline for clinical radiology data, transforming unstructured multi-study patient timelines into structured JSON for downstream survival analytics
- Implemented self-healing loops (generate → validate → repair) to handle LLM hallucination and formatting failures in real-world medical text
- Trained transformer-based survival models using Cox proportional hazards loss for time-to-event prediction. Developed evaluation pipelines using concordance index (C-index) and ranking-based survival metrics

Student Researcher — AI Healthcare Intelligence & Clinical Analytics - UofM & Trinity Health Jan 2025 – Dec 2025

- Processed 200K+ medical billing and clinical progress notes; engineered NLP features using TF-IDF and Word2Vec
- Built end-to-end ML pipelines (XGBoost, Random Forest, SVM, Logistic Regression) achieving 82% F1-score for billing compliance classification
- Developed LLM-assisted auditing workflows using Azure OpenAI and Microsoft Fabric for anomaly detection
- Reduced manual audit review time by 30% by integrating model outputs into production analytics pipelines

Data Analysis Intern - Haier Global Business Jun 2024 – Aug 2024

- Analyzed 500K+ e-commerce transactions using Python and SQL to identify pricing inefficiencies and growth opportunities
- Built automated data pipelines and dashboards reducing analysis time by 40%;
- Delivered pricing insights adopted by product managers contributing to 58% YoY sales growth.

Alfred — AI Chief of Staff Feb 2026 – Present

- Designed and built a mobile-first AI Chief of Staff that transforms emails, calendars, and conversations into commitments, priorities, and executable actions.
- Built an agentic workflow integrating Gmail, Calendar, LLM-based commitment extraction, deterministic priority ranking, and human-approved execution.
- Implemented permission-gated ActionProposal architecture with audit logs and transparent execution to ensure safe real-world automation.
- Designed explainable decision pipelines that separate understanding, prioritization, planning, and execution instead of relying solely on LLM generation.

PulsePilot — Personalized AI Behavioral Intelligence Platform

- Built a personalized AI platform that converts natural-language workout, meal, and mood logs into structured behavioral records.
- Designed rolling personal baselines and statistical trend analysis to identify individualized behavior patterns rather than population-level recommendations.
- Developed explainable AI insights linking exercise, nutrition, stress, consistency, and mood through interpretable analytics.
- Built end-to-end data pipelines supporting natural-language logging, behavioral analysis, and interactive trend visualization.

U.S. Equity Research Agent

- Built a multi-agent AI system that transforms fragmented market information into structured daily trading research.
- Designed specialized agents for macroeconomics, company analysis, technical signals, and news synthesis.
- Implemented evidence validation, thesis comparison, and deterministic decision workflows to produce reviewable trading plans.
- Developed morning planning and evening review pipelines enabling continuous feedback and transparent decision making.